PRIVACY-PRESERVING FEDERATED LEARNING FRAMEWORK FOR MULTI-HOSPITAL MEDICAL DATA ANALYSIS

CKD Case Study - Prediction Report

Academic research prototype only. This is not a medical diagnosis.

1. Patient Prediction

Feature	Value
Age	50
Bp	80
Sg	1.02
Al	1
Su	0
Rbc	normal
Pc	normal
Pcc	notpresent
Ba	notpresent
Bgr	100
Bu	40
Sc	1.2
Sod	140
Pot	4.5
Hemo	13
Pcv	40
Wbcc	8000
Rbcc	5
Appet	good
Prediction	CKD
Model-estimated CKD probability	81.88%

2. Federated Learning Results

These are project-level experiment results, not results for this individual patient.

Final Performance

Method	Accuracy	Precision	Recall	F1	ROC-AUC
Normal FL	0.592	0.5477	0.9981	0.7073	0.5928
Fixed-DP FL	0.5617	0.5299	0.9981	0.6922	0.5554

Method	Accuracy	Precision	Recall	F1	ROC-AUC
Proposed Adaptive-DP FL	0.5099	0.5857	0.0265	0.0507	0.5339

Hospital-Wise Results

Method	Hospital	Accuracy	Precision	Recall	F1	ROC-AUC
Normal FL	Hospital 1	0.625	0.625	1.0	0.7692	0.33
Normal FL	Hospital 2	0.6073	0.5544	0.9986	0.713	0.616
Normal FL	Hospital 3	0.5734	0.5339	0.9973	0.6954	0.5839
Normal FL	Hospital 4	0.65	0.65	1.0	0.7879	0.5
Fixed-DP FL	Hospital 1	0.5875	0.6104	0.94	0.7402	0.2423
Fixed-DP FL	Hospital 2	0.5734	0.5338	1.0	0.696	0.578
Fixed-DP FL	Hospital 3	0.5462	0.5183	1.0	0.6828	0.5458
Fixed-DP FL	Hospital 4	0.65	0.65	1.0	0.7879	0.5
Proposed Adaptive-DP FL	Hospital 1	0.525	0.5857	0.82	0.6833	0.4437
Proposed Adaptive-DP FL	Hospital 2	0.5116	0.0	0.0	0.0	0.5297
Proposed Adaptive-DP FL	Hospital 3	0.5116	0.0	0.0	0.0	0.5392
Proposed Adaptive-DP FL	Hospital 4	0.35	0.0	0.0	0.0	0.5

Missingness and Adaptive Noise

Hospital	Missing Level	Adaptive Noise
Hospital 1	0.05	0.05
Hospital 2	0.15	0.1
Hospital 3	0.25	0.2
Hospital 4	0.4	0.3

Normal Federated Learning

Round	Accuracy	Precision	Recall	F1	ROC-AUC
1.0	0.5412	0.5184	1.0	0.6829	0.5138
2.0	0.5655	0.532	1.0	0.6945	0.4893
3.0	0.5859	0.544	1.0	0.7046	0.5923
4.0	0.5843	0.543	1.0	0.7038	0.5916
5.0	0.5843	0.543	1.0	0.7038	0.5848
6.0	0.5837	0.5426	1.0	0.7035	0.5828
7.0	0.5827	0.5421	1.0	0.703	0.589
8.0	0.592	0.5477	0.9981	0.7073	0.5928

Fixed-DP Federated Learning

Round	Accuracy	Precision	Recall	F1	ROC-AUC
1.0	0.4939	0.4939	1.0	0.6612	0.4536
2.0	0.4939	0.4939	1.0	0.6612	0.5462

Round	Accuracy	Precision	Recall	F1	ROC-AUC
3.0	0.5224	0.5084	1.0	0.6741	0.5672
4.0	0.5546	0.5259	1.0	0.6893	0.5033
5.0	0.5665	0.6886	0.2232	0.3371	0.5972
6.0	0.4585	0.1629	0.0233	0.0407	0.517
7.0	0.508	0.875	0.0045	0.009	0.5722
8.0	0.5617	0.5299	0.9981	0.6922	0.5554

Adaptive-DP Federated Learning

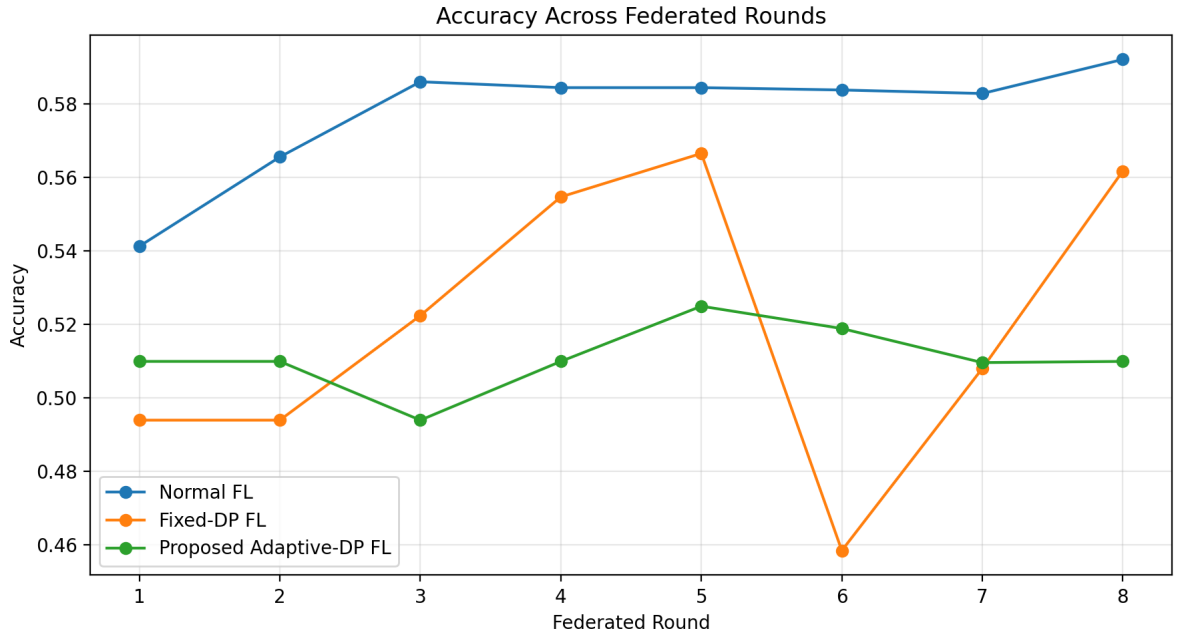
Round	Accuracy	Precision	Recall	F1	ROC-AUC
1.0	0.5099	0.5857	0.0265	0.0507	0.5381
2.0	0.5099	0.5857	0.0265	0.0507	0.5572
3.0	0.4939	0.4939	1.0	0.6612	0.4652
4.0	0.5099	0.5857	0.0265	0.0507	0.5287
5.0	0.5249	0.585	0.1313	0.2145	0.5464
6.0	0.5188	0.507	0.9379	0.6582	0.5224
7.0	0.5096	0.5018	1.0	0.6683	0.4459
8.0	0.5099	0.5857	0.0265	0.0507	0.5339

Membership Inference Attack Results

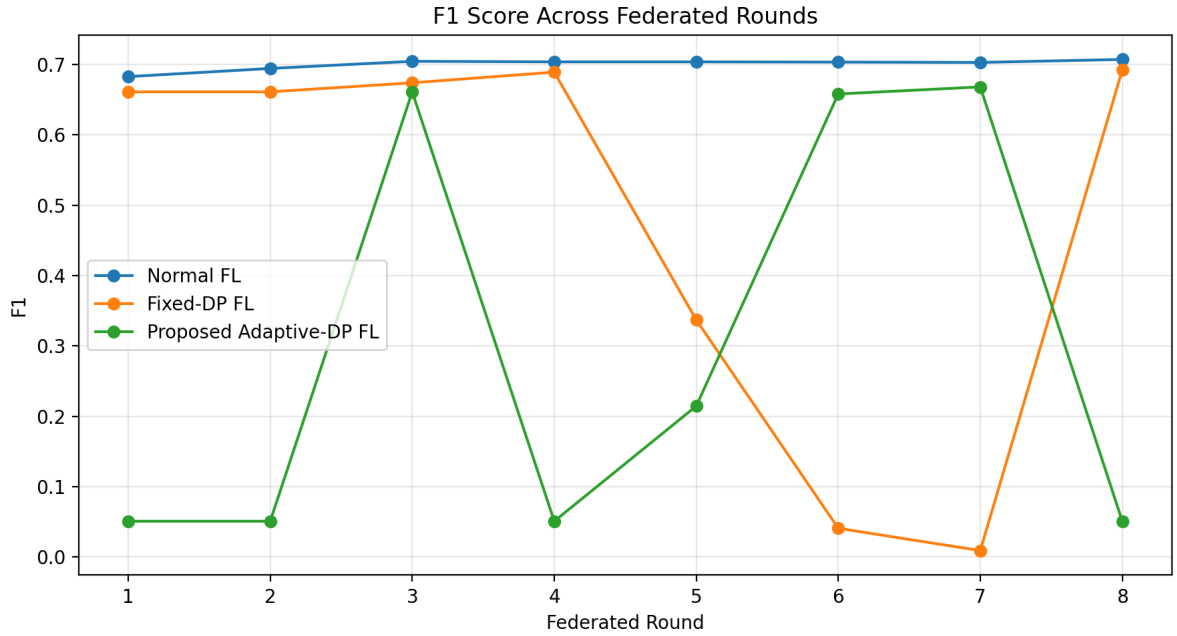
Method	MIA Accuracy	MIA AUC	MIA Threshold
Normal FL	0.5056	0.5066	0.8766
Fixed-DP FL	0.5054	0.5037	0.7451
Proposed Adaptive-DP FL	0.5021	0.5019	1.0

3. Project Graphs

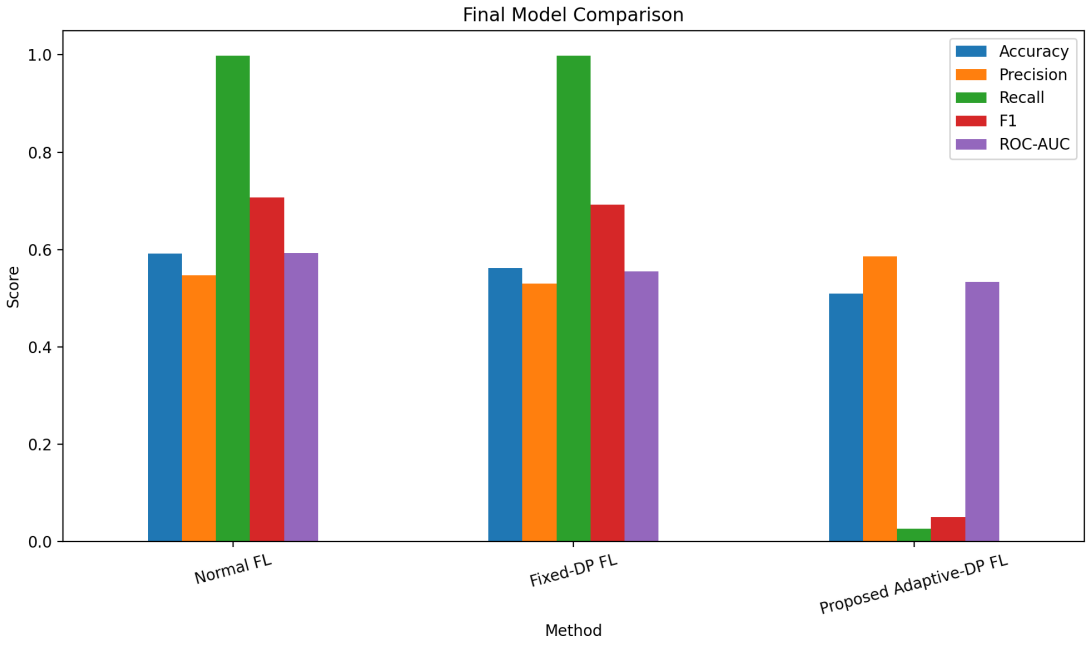
Accuracy Across Rounds



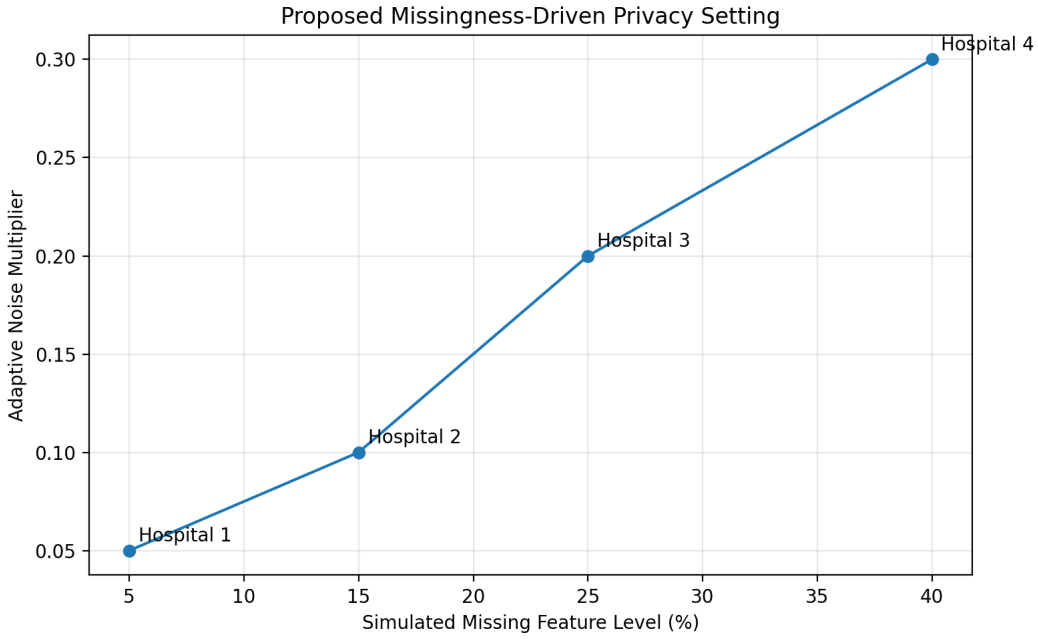
F1 Across Rounds



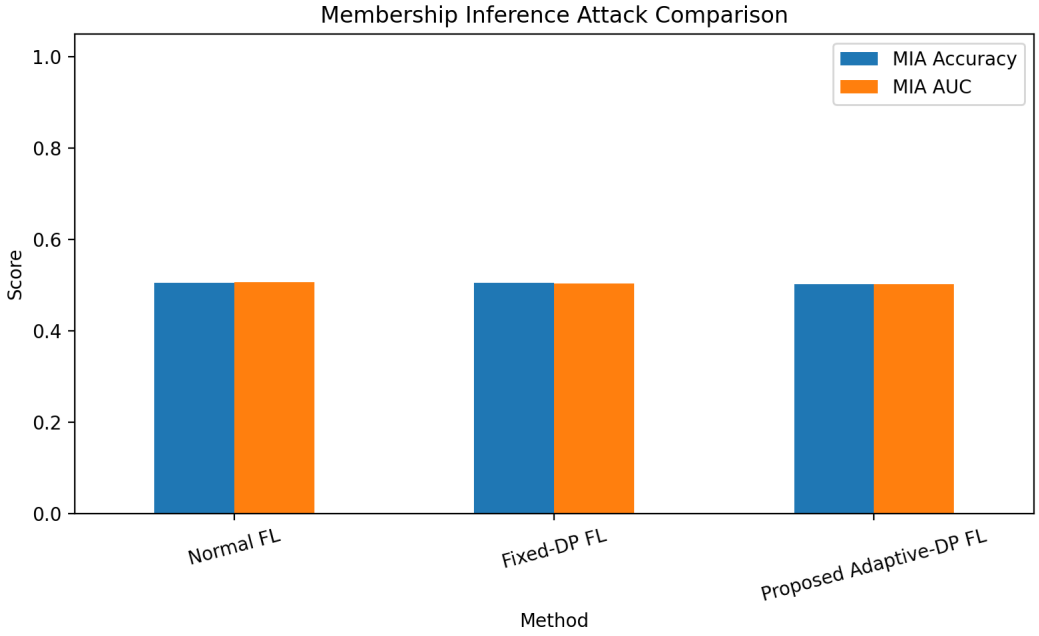
Final Metric Comparison



Missingness vs Adaptive Noise



Membership Inference Attack Comparison



Adaptive-DP Confusion Matrix

